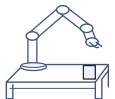


Source organization

Successful simulation replays

Fig. 2

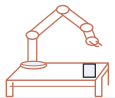
Successful replays (simulation)



- 50 episodes per task
- Current states actions A_{ref}

State records

Reference actions (same episodes)



- 400-step episodes
- 14 action channels actions A_{ref}

Action records

Preprocessing (build training records)

Verify source ID

40 / 10
train / dev

Build chunks

Training record
(same task)

A_{ref}

Reference actions

S_0

(initial replay layout)

Policy-error calibration

Calibrate within each task

Fig. 3

Episode-separated fitting (five auxiliary policies)

Excluded fold: 8



o_t
 Y_t
...

5-fold episode split
(per task)

Fit supervision



o_t
 Y_t

Fitting folds: 32



o_t
 Y_t
...

Fit on 32
(o_t, Y_t)

Standardized MSE

MSE loss
(fit folds only)

Out-of-episode calibration

Inputs: excluded fold
Policy fitted on the other 32

Fit-fold updates only

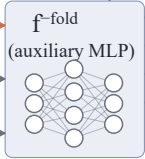
$$E_t = \hat{A}_t - Y_t$$

(raw units)

$q_t \cdot 14\text{-D}$
(robot state)

$p_t \cdot 3/6\text{-D}$
(object position)

$t/T \cdot 1\text{-D}$
(elapsed time)



$f = 1$
 $f = 2$
...
 $f = 5$

Pool 5 folds by time
1960 chunks / task

Errors
 $\{E_t\}$

A_{ref} (reference sequence for replay)

s_0 (reset the same initial layout for each attempt)

Recovery supervision

Replay and record

Fig. 4

Bounded perturbation (8 steps)

$E_{1:8}$

$g = 0$

Gripper



RMS scale



Cap $\pm 3\sigma$

+ Add pulse

Replay

Replay audit

(all trials)

A_{exec}
executed
source

δ_t
observed
endpoint

r_{399}
final reward
seed / scale



Recovery windows

$t \geq u$

Replay state + executed action



Recovery record

(window)

Replay state

δ_t (current)

$\oplus p$

e

$\uparrow q$

t/T



$Y_t = A_{exec, t:t+16}$

(reference tail)



source $\cdot u \cdot t \cdot \text{seed} \cdot \text{scale}$



Action-chunk policy

1:1 original + recovery